Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 24-01-2026**
Time: 09.00Am to 12.00Pm **GTM-18** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**1**	2	**3**	3	**1**	4	**3**	5	**1**
6	**4**	7	**3**	8	**1**	9	**3**	10	**1**
11	**1**	12	**2**	13	**3**	14	**4**	15	**2**
16	**4**	17	**4**	18	**4**	19	**1**	20	**3**
21	**70**	22	**109**	23	**1**	24	**3**	25	**7**

PHYSICS

26	**2**	27	**3**	28	**3**	29	**4**	30	**1**
31	**2**	32	**2**	33	**2**	34	**1**	35	**1**
36	**3**	37	**2**	38	**1**	39	**3**	40	**4**
41	**3**	42	**2**	43	**2**	44	**3**	45	**3**
46	**8**	47	**10**	48	**20**	49	**6**	50	**5**

CHEMISTRY

51	**4**	52	**4**	53	**1**	54	**4**	55	**2**
56	**3**	57	**2**	58	**1**	59	**3**	60	**3**
61	**1**	62	**1**	63	**3**	64	**3**	65	**1**
66	**4**	67	**2**	68	**2**	69	**4**	70	**3**
71	**7**	72	**104**	73	**6**	74	**2**	75	**2**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

1. (A) Put $x = \sin\theta$

$\therefore \int_0^{\pi/2} \frac{d\theta}{1+\cos\theta} = 1$

(B) $f'(x) = 5x^4 + \frac{1}{2}e^{x/2} + 3x^2e^{x^3} > 0$

$\therefore f$ is one-one map

$gf(x) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$

$g(f(x)) = \frac{1}{f'(x)}$

& $f(0) = 2 \ \ \therefore g'(2) = 2$

(C) $T_r = \frac{1}{\sqrt{n}\sqrt{r}} = \frac{1}{n\left(\sqrt{r/n}\right)}$

$\therefore$ Sum $= \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{4n} \frac{1}{\sqrt{r/n}} = \int_0^4 \frac{1}{\sqrt{x}} dx = 4$

(D) $f'(x) = \left[(2x)^3 - 2(2x) + 4\right] \times 2 \ - \left[(x-1)^3 - 2(x-1) + 4\right]$

2. If $0 < a < b < c < d$

$D = [1 - a(b+c)]^2 - 4a[abc - d]$

$= 1 - 2a(b+c) + a^2\left[(b+c)^2 - 4bc\right] + 4ad$

$= [1 - a(b-c)]^2 + 4a(d-c) > 0$ by (1)

$\therefore$ roots are real and distinct.

$f(c) = ac^2 + \left[1 - a(b+c)\right]c + abc - d$

$= c - d < 0 (\because < d)$

Also, $f(d) = ad^2 + \left[1 - a(b+c)\right]d + -abc - d$

$= ad(d-c) - ab(d-c)$

$= a(d-c)(d-b) > 0$ by (1)

$\because f(c)$ and $f(d)$ are of opposite signs, one of the real roots lies between c and d

3. A) $\frac{T_7}{T_6} > 1 \quad \Rightarrow \left(\frac{n-6+1}{6}\right)\left(\frac{2}{3}\right) > 1$

$\Rightarrow n > 14$

similarly $\frac{T_7}{T_8} > 1$

$nn < \frac{33}{2} \Rightarrow n = 16$

as there is only are middle term

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$\Rightarrow$ **n** must be even $\Rightarrow$ $\mathbf{n}=16$

$n=4.3+4\Rightarrow\lambda=3,\mu=4$

(B) $(1+x)^{101}\left(1+x^2-x\right)^{100}$

$=(1+x)\left\{(1+x)\left(1+x^2-x\right)\right\}^{100}$

$=(1+x)\left(1+x^3\right)^{100}$

$=(1+x)\left(1+{}^{100}C_1x^3+\ldots\ldots+{}^{100}C_{100}x^{300}\right)$

$=1+{}^{100}C_1x^3+{}^{100}C_2x^6+{}^{100}C_3x^9$

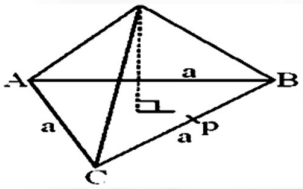
$+\ldots\ldots\ldots+{}^{100}C_{100}x^{300}$

$+x+{}^{100}C_1x^4+{}^{100}C_2x^7+\ldots\ldots+$

${}^{100}C_{100}x^{301}$

$\mathbf{n}=1+100+101=202$

4.



Let a point **p** lies on midpoint **BC**;

$$P(1+\lambda,2+2\lambda,-\lambda)$$

$$\overrightarrow{AP}\cdot\overrightarrow{BC}=0$$

$$\left((1+\lambda)\hat{i}+(2+2\lambda)\hat{j}-\lambda\hat{k}\right)\cdot\left(\hat{i}+2\hat{j}-\hat{k}\right)=0$$

$$\therefore\lambda=\frac{-5}{6}$$

$$\therefore p=\left(\frac{1}{6},\frac{2}{6},\frac{5}{6}\right)$$

$$\frac{\sqrt{3}}{2}\cdot a=AP=\frac{1}{6}\sqrt{30}\Rightarrow a=\frac{\sqrt{10}}{3}$$

Normal vector of plane ABC is

$$\vec{n}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&-1\\1&2&0\end{vmatrix}=2\hat{i}-\hat{j}$$

Plane ABC is $2x-y=0$

5. CONCEPTUAL

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6. $S_3 = \dfrac{\alpha^3+1}{\alpha^5-\alpha^4-\alpha^3+\alpha^2} + \dfrac{\beta^3+1}{\beta^5-\beta^4-\beta^3+\beta^2}$

$$\frac{\alpha(\alpha+3)+1}{\alpha^2(\alpha-1)(\alpha^2-1)} = \frac{4\alpha+4}{(\alpha+3)(\alpha-1)(\alpha+2)}$$

$$= \frac{4\alpha+4}{3\alpha(\alpha+2)} = \frac{4}{9}$$

7. We have

$$f_1(x) = 2\sum_{r=1}^{n} \frac{\sin\left(x+\frac{r\pi}{6}\right)-\left(x+(r-1)\frac{\pi}{6}\right)}{\cos\left(x+(r-1)\frac{\pi}{6}\right)\cdot\cos\left(x+\frac{r\pi}{6}\right)}$$

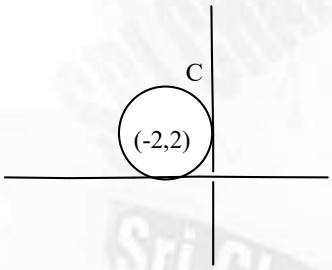
$$= 2\left[\left(\tan\left(x+\frac{\pi}{6}\right)-\tan x\right)+\left(\tan\left(x+\frac{2\pi}{6}\right)-\tan\left(x+\frac{\pi}{6}\right)\right)\right.$$

$$+\left(\tan\left(x+\frac{3\pi}{6}\right)-\tan\left(x+\frac{2\pi}{6}\right)\right)+\ldots+$$

$$\left.\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan\left(x+(n-1)\frac{\pi}{6}\right)\right)\right]$$

$$\Rightarrow f_1(x) = 2\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan x\right)$$

8.



$C:(x+2)^2+(y-2)^2=2^2$

$C_1:(x-2)^2+(y-5)^2=r^2$

If both circle intersect at two points

$\therefore |r_1-r_2| < c_1c_2 < r_1+r_2$

$\Rightarrow |r_1-r_2| < 5 < 2+r$

$\Rightarrow 3 < r < 7 \Rightarrow r \in (3,7)$

$\therefore \alpha = 3, \beta = 7, So, 3\beta - 2\alpha = 15$

9. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \Rightarrow x^{2020}e^x = \sum \frac{x^{n+2020}}{n!}$ Now diff

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$$\Rightarrow \left(2020x^{2019}+x^{2020}\right)e^x=\sum(n+2020)\frac{x^{n+2019}}{n!}$$ Multiply by x

$$\left(2020x^{2020}+x^{2021}\right)e^x=\sum\frac{(n+2020)x^{n+2020}}{n!}$$

$$\left(2020^2x^{2019}+2021x^{2020}\right)e^x+e^x\left(2020x^{2020}+x^{2021}\right)$$

$$e^x\left(2020^2x^{2019}+4041x^{2020}+x^{2021}\right)=\sum\frac{(n+2020)^2x^{n+2019}}{n!}$$

10. $p(x)=ax^4+bx^3$

We get $a=-6, b=-2$

11. CONCEPTUAL

12. Given $\frac{d}{dx}\left(\frac{f^1(x)}{e^x}\right)$ $\therefore f^1(x)=e^x(x+1)$

13. $XX^T=I$,

$PQ=AXB^2X^TA$

$=3(AXX^TA)$

$=3A^2$

$=\begin{pmatrix}12 & 0\\ 0 & 27\end{pmatrix}$

14. $\left(\frac{1}{a_1}-\frac{1}{a_2}\right)+\left(\frac{1}{a_2}-\frac{1}{a_3}\right)+\ldots\left(\frac{1}{a_9}-\frac{1}{a_{10}}\right)=\frac{9d}{64}$

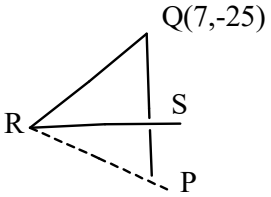
$$\Rightarrow \frac{a_{10}-a_1}{a_1a_{10}}=\frac{9d}{64}$$

$$\Rightarrow a_1a_{10}=64$$

Also $(a_1+a_{10})\left(\frac{1}{a_1a_{10}}+\ldots\frac{1}{a_{10}a_1}\right)=\frac{a_1+a_{10}}{10}\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)\Rightarrow 2\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)=\frac{a_1+a_{10}}{10}\left(\frac{1}{a_1}+\ldots\frac{1}{a_{10}}\right)$

15. (R) is (5,5,8)

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$$ar(\Delta PQR)=\left(2\frac{1}{2}(QS)(RS)\right)^2$$

16. Chord of contact of R $hx+2ky=6$

compare with $\frac{x}{2}\cos\theta+y\sin\theta=1$ $\qquad \frac{2h}{\cos\theta}=\frac{2k}{\sin\theta}=6$ $\qquad h^2+k^2=9$

which is director circle of $x^2+2y^2=6$.

17. Equation of the tangent at $\mathbf{P}(5\cos\theta,4\sin\theta)$ is

$(4\cos\theta)x+(5\sin\theta)y-20=0$

Equation of the tangent at $P(5\sin\theta)x-(4\sin\theta)\,y-\sin\theta\cos\theta=0$

$$CM=\left|\frac{0+0-20}{\sqrt{16\cos^2\theta+25\sin^2\theta}}\right| \text{ and } CN=\left|\frac{0-0-9\sin\theta\cos\theta}{\sqrt{25\sin^2\theta+16\cos^2\theta}}\right|$$

Area of the rectangle $CMPN=CN\times CN=\frac{|180\sin\theta\cos\theta|}{16\cos^2\theta+25\sin^2\theta}$

Here $\mathbf{P}=\left(\frac{5}{\sqrt{2}},2\sqrt{2}\right)\equiv(5\cos\theta,4\sin\theta)$

18. $$\sqrt{10}=\frac{\begin{vmatrix}2&-1&2\\1&m&0\\\frac{1}{2}&1&-4\end{vmatrix}}{\left|\sqrt{4m^2+4+(2m+1)^2}\right|}$$

$$\sqrt{10}=\left|\frac{9m+2}{\sqrt{8m^2+4m+5}}\right| \quad 10\left(8m^2+4m+5\right)=(9m+2)^2$$

19. Suppose that the first term of the sequence ΔA is d then

$\Delta A=\{d,d+1,d+2,\ldots\ldots(d+(n-1))\}$

Hence $A=(a_1,a_1+d,a_1+d+(d+1))$,

$a_1+d+(d+1)+(d+2)\ldots\ldots)$

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$a_n = a_1 + (n-1)d + \frac{1}{2}(n-1)(n-2)$

so a_n is a quadratic polynomial in n.

so $a_n = \frac{(n-\alpha)(n-\beta)}{2}$

Since $a_{19} = a_{92} = 0$ we must have $a_n = \frac{1}{2}(n-19)(n-92)$

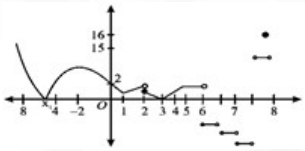
20. Solve, $[2x]\operatorname{sgn}\left(x - \frac{15}{2}\right) - 12, 6 \le x < \frac{13}{2} - 13, \frac{13}{2} \le x < 7$

$-14, 7 \le x < \frac{15}{2}\ 0, x = \frac{15}{2}\ 15, \frac{15}{2} < x < 8\ 16\ ,\ x = 8$

Discontinuous at $x = 2, 6, \frac{13}{2}, 7, \frac{15}{2}, 8, \therefore \alpha = 6$

Non-differentiable at $x = x_1, 0, 1, 2, 3, 5, 6, \frac{13}{2}, 7, \frac{15}{2}, 8, (x_1$ be a root of $x^2 + 5x - 2 = 0$ in $x < 0)$

$\therefore \beta = 11$



21. As the deviation from max sum is 4

22. dot with $\mathrm{B} \times \mathrm{C}$

$(\vec{\mathrm{B}} \times \vec{\mathrm{C}}) \cdot (\vec{\mathrm{B}} \times \vec{\mathrm{C}}) = \mathrm{x}\left[\vec{\mathrm{A}}\ \ \vec{\mathrm{B}}\ \ \vec{\mathrm{C}}\right]$

$x = \frac{(\vec{B} \times \vec{C})^2}{\left[\vec{A}\ \ \vec{B}\ \ \vec{C}\right]};\ x = \frac{\vec{B}^2\vec{C}^2 - (\vec{B} \cdot \vec{C})^2}{\left[\vec{A}\ \ \vec{B}\ \ \vec{C}\right]}$

dot with $\vec{C} \times \vec{A}$ gives $y = \frac{(\vec{B} \times \vec{C}) \cdot (\vec{C} \times \vec{A})}{\left[\vec{A}\vec{B}\vec{C}\right]}$

and dot with $\vec{A} \times \vec{B}$ gives $z = \frac{(\vec{B} \times \vec{C}) \cdot (\vec{A} \times \vec{B})}{\left[\vec{A}\vec{B}\vec{C}\right]}$

$$\left[\vec{\mathrm{A}}\ \ \vec{\mathrm{B}}\ \ \vec{\mathrm{C}}\right] = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 1(1+1) - 2(-2-3) = 12$$

$\vec{\mathrm{B}}^2\vec{\mathrm{C}}^2 - (\vec{\mathrm{B}} \cdot \vec{\mathrm{C}})^2 = (6)(2) - (0) = 12$

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23. $$\frac{1}{abc}\begin{vmatrix} -abc & a(b^2+bc) & a(c^2+bc) \\ b(a^2+ac) & -abc & b(c^2+ac) \\ c(a^2+ab) & c(b^2+ab) & -abc \end{vmatrix} = \begin{vmatrix} -bc & a(b+c) & a(b+c) \\ b(a+c) & -ac & b(a+c) \\ c(a+b) & c(a+b) & -ab \end{vmatrix}$$

$$= \begin{vmatrix} -bc & ab+bc+ac & ab+bc+ac \\ ab+bc & -(ab+bc+ac) & 0 \\ bc+ac & 0 & -(ab+bc+ac) \end{vmatrix}$$

$$C_2 \to C_2 - C_1 \,\&\, C_3 \to C_3 - C_1 = (ab+bc+ac)^2 \begin{vmatrix} -bc & 1 & 1 \\ ab+bc & -1 & 0 \\ bc+ac & 0 & -1 \end{vmatrix}$$

24. $$f(x) = \begin{cases} ax^3+bx^2 & ; \quad x^2<1 \quad \Rightarrow -1<x<1 \\ \frac{1}{x} & ; \quad x^2>1 \quad \Rightarrow x<-1 \text{ or } x>1 \\ \frac{a+b+1}{2} & ; \quad x=1 \end{cases}$$

$\because f(x)$ is continuous at $x=1 \Rightarrow a+b=1$ and $f(x)$ is continuous at $x=-1 \Rightarrow b-a=-1$

$\therefore a=1,\ b=0 \ \ \therefore\ A(-1,3)$ and $B(1,-1) \qquad \therefore g'(x)=\lambda(x-1)(x+1)$

25. $$(3ff''+f\cdot f''')(x) = \left(\left(ff''+(f')^2\right)(x)\right)'$$

$$\left(f''+(f')^2\right)(x) = \left((ff')(x)\right)'$$

$$\therefore (3ff''+f''')(x) = \left(f(x)\cdot f'(x)\right)''$$

Min. number of roots of $f(x)=5$

Min. number of roots of $f'(x)=4$

$\Rightarrow$ Min. number of roots of $f(x)\cdot f'(x)=5+4=9$

$\therefore$ Min. number of roots of $\left(f(x)\cdot f'(x)\right)''=7$

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PHYSICS

26. Let θ be the angle formed by the non horizontal section of the rope and the horizontal. Let also m be the mass of the box and T the tension in the rope. Then T is also the force applied on the rope by Arnav. The vertical component of the force applied to the box is T = T sin θ, while its horizontal component is T = T cosθ. Therefore, the force of friction is

$F = \mu(mg - T) = \mu(mg - T\sin\theta)$.

Let us now consider all available answers one by one.

A) Let us determine Lily's force T so that its horizontal component matches the kinetic friction for any angle θ :

$T = F \Rightarrow T\cos\theta = \mu(mg - T\sin\theta)$

$\Rightarrow T = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$

Because θ changes over time, force T cannot be constant, so this answer choice is out.

B) From the expression $F = \mu(mg - T\sin\theta)$ we see that as θ increases over time, sin θ does too, and so F decreases over time. This is the **correct answer**.

C) The normal force of the ground is $N = mg - T = mg - T\sin\theta = mg(1 - \frac{\mu}{\cot\theta + \mu})$. As θ increases over time, cot θ decreases, and so N will decrease over time. This answer is out.

D) The pressure is proportional to N, so it decreases too. This answer is out.

27. When ball is inserted in sphere, potential of ball becomes 10001 volt and potential of outer sphere is more than 10000 V but less than 10001 V. and positive charge flow from higher voltage to lower voltage.

28. Gravitation field inside a uniform Sphere is $g = \frac{GMr}{R^3}$, as due to expansion mass has not changed bur radius R has becomes 2R and Distance from center r is same so field will become 1/8 of initial.

29. We can determine the speed v of the bead as it leaves the wire by using the conservation of energy: ⇒ $v = \sqrt{4gR}$

Now bead will do horizontal projectile motion and range $D = v\sqrt{\frac{2H}{g}} = \sqrt{8Rh}$

30. CONCEPTUAL

31. CONCEPTUAL

32. CONCEPTUAL

33.

Here $J = \frac{I}{\pi R^2 - \frac{\pi R^2}{4}} = \frac{4I}{3\pi R^2}$

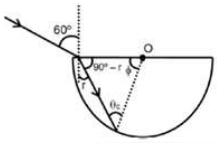
$B_{net} = \frac{\mu_0 J}{2}\cdot\frac{R}{2} - \frac{\mu_0 J(R/2)^2}{2R} = \frac{\mu_0 JR}{8}$

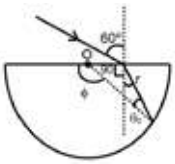
Putting the value of J we get $B_{net} = \frac{\mu_0 I}{6\pi R}$

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34.

Rays left of O Rays right of O





$\sin 60° = \sin r \times 2$

$\frac{\sqrt{3}}{2} = 2\sin r$

$\therefore r = \sin^{-1}\frac{\sqrt{3}}{4} = 25°$

$\phi = 180 - [90° - 25° + 30°] = 85°$

35. If E_0 is magnitude of electric field then

$$\frac{1}{2}\varepsilon_0 E^2 \times C = I, E_0 = \sqrt{\frac{2I}{C\varepsilon_0}}, B_0 = \frac{E_0}{C}$$

Direction of $\vec{E} \times \vec{B}$ will be along $+\hat{j}$.

36. Velocity of efflux $= \sqrt{2gh}$

Range $= \sqrt{2gh} \times \sqrt{\frac{2(H-h)}{g}} = \sqrt{4\left(Hh - h^2\right)}$

Time empty container $= \frac{A}{A_0}\sqrt{\frac{2H}{g}}$

37. $4mg - 2F = 4\text{ma}$

$3mg - F - T = 3\text{ma}$

$T = \text{mg} - F - 3\text{ma} = 3mg - F - 3m\left(\frac{4\text{mg} - 2F}{4\text{m}}\right)$

$= 3mg - F - 3mg + \frac{3F}{2} = \frac{F}{2}$ $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{F}{2\mu}}$ $t = \frac{\ell}{v} = \ell\sqrt{\frac{2\mu}{F}}$

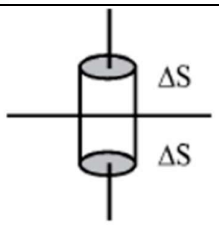
38. $\frac{1}{f_L} = \frac{\frac{3}{2}-1}{-10} + \frac{\frac{4}{3}-\frac{3}{2}}{+15} + \frac{1-\frac{4}{3}}{-15} = \frac{-1}{20} - \frac{1}{90} + \frac{1}{45} = \frac{-7}{18}$

$\frac{1}{f_q} = \frac{1}{f_m} - \frac{-2}{f_L} = \frac{-2}{15} + \frac{7}{90} = \frac{-5}{90} = f_{q=-18cm}$

39. $a = \mu g, \alpha = \frac{\mu mgR}{mR^2/2} = 2\frac{\mu g}{R}, V = Rw, \mu gt = R\left[w_0 - 2\frac{\mu g}{R}t\right], t = \frac{w_0}{3\mu g}$

40.

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$E_1 = \frac{\sigma}{K_1\varepsilon_0}$ $E_2 = \frac{\sigma}{K_2\varepsilon_0}$

$E_2\Delta S - E_1\Delta S = \frac{\sigma\Delta S}{\varepsilon_0}$ $\frac{\sigma}{K_2\varepsilon_0} - \frac{\sigma}{K_1\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$

$\sigma\left[\frac{1}{K_2} - \frac{1}{K_1}\right] = \sigma$ $\sigma\left[\frac{1}{6} - \frac{1}{2}\right] = \sigma$ $\sigma\left[\frac{2-6}{12}\right] = \sigma$ $-\frac{\sigma}{3} = \sigma$

41.

$m.\frac{dv}{dt} = -6\pi\eta rv$

$\frac{dv}{dt} = -\frac{v}{2}$

$\frac{dv}{v} = \frac{-dt}{2}$

$-\int_2^1 \frac{dv}{v} = \frac{1}{2}\int_{t_1}^{t_2} dt$

$-[1n\,v]\frac{1}{2} = \frac{1}{2}\Delta t$

$\text{In}\,2 = \frac{\Delta t}{2}$

$2\,\text{In}\,2 = \Delta t$

42. CONCEPTUAL

43. CONCEPTUAL

44. For decay $X^{230} \rightarrow Y^{226} + He^4 + Q$

$K_a = \left(\frac{A-4}{A}\right)Q \Rightarrow Q = \frac{6.8\times 230}{226} = 6.92\text{MeV}$

For $X^{230} \rightarrow {}^*Y^{226} + He^4$

$\sqrt{2\,m_y\,K_y} = \sqrt{2\,m_a\,K_a} \Rightarrow K_y = \frac{4\times 5.2}{226} = 0.09\text{MeV}$

$E_Y = Q - K_a - K_y = 6.92 - (5.2 + 0.09)$

$= 1.63\text{MeV}$

45. For polytrophic process $PV^m =$ constant

$C = C_v + \left(\frac{R}{1-m}\right)\left(\text{Here } m = \frac{3}{2} \text{ and } C_v = \frac{3R}{2}\right)$

$\Rightarrow C = \frac{3R}{2} + \frac{R}{1-\frac{3}{2}} = \frac{3R}{2} - 2R = -0.5R$

$Q = nC\Delta T = 1(-0.5R)(-26) = +13R$

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46. when the man is at rest, all of his initial potential energy was converted to elastic potential energy in the cord, so $mgH = \frac{1}{2} kh^2$
since the maximum tension is four times the man's weight, $4mg = kh$

47. CONCEPTUAL

48. **For series** $L_{eq} = L_1 + L_2$

For parallel $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$

49.

$$I = \frac{M}{2}\left(R_1^2 + R_2^2\right)$$

$$dI = \frac{M}{2}\left[2R_1 dR_1 + 2R_2 dR_2\right] + \frac{dM}{2}\left[R_1^2 + R_2^2\right]$$

$$\frac{dI}{I} \times 100 = 5.8\%$$

50. CONCEPTUAL

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CHEMISTRY

51. $N_2O_5(g) \rightleftarrows 2NO_2(g) + \frac{1}{2}O_2(g)$

Initially: 600 torr

$(600-p)torr \quad 2p\,torr \quad \frac{p}{2}torr$

At equilibrium:

Total press is $\left(600+\frac{3p}{2}\right)torr$

$\therefore 600+\frac{3p}{2}=960 \; or \; \frac{3p}{2}=360 \; or \; p=240\,torr$

$\therefore$ fraction of $N_2O_5(g)$ decomposed $=\frac{240}{600}=0.4$

52. NO_2^- is ambidentate ligand

53. wt of acetic acid $=2.05\times60=123$gm

wt of 1000 ml solution = 1020 gm

wt of H_2O = 1020 – 123 = 897 gm

$n_{CH_3COOH}=\frac{123}{60}=2.05$ $\qquad$ $n_{H_2O}=\frac{897}{18}=49.83$

n = 2.05 + 49.83 = 51.88

$\chi_{H_2O}=\frac{49.83}{51.88}=0.96$

54. Value of equilibrium constant changes with (a) Temperature, (b) mode of representation of reaction only

55. $P_A=P_A^o x_A=200\times\frac{1}{2}=100$

$P_B=P_B^o x_B=100\times\frac{1}{2}=50.$

$P_S=150 \qquad P_A=P_S.Y_A \qquad 100=150.Y_A \qquad Y_A=0.66$

56. $Hg(l)+Cl_2(g)\rightarrow HgCl_2(s),$

$rate\,of\,reaction=-\frac{\Delta[Hg]}{\Delta t}=-\frac{\Delta[Cl_2]}{\Delta t}=\frac{\Delta[HgCl_2]}{\Delta t}$

$2HI(g)\rightarrow H_2(g)+I_2(g),$

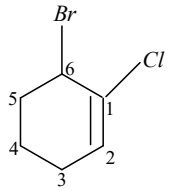
$rate\,of\,reaction=-\frac{1}{2}\frac{\Delta[HI]}{\Delta t}=\frac{\Delta[H_2]}{\Delta t}=\frac{\Delta[I_2]}{\Delta t}$

$5Br^-(aq)+BrO_3^-(aq)+6H^+(aq)\rightarrow 3Br_2(aq)+3H_2O(l)$

$rate\,of\,reaction=-\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t}=-\frac{\Delta[BrO_3^-]}{\Delta t}=-\frac{1}{6}\frac{\Delta[H^+]}{\Delta t}=+\frac{1}{3}\frac{\Delta[Br_2]}{\Delta t}=+\frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$

57.

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58. All the hydrides of group V elements have one lone pair of electrons on their central Atom. Therefore, they can act as Lewis bases. The basic character of these hydrides decrease down the group.

59. Conceptual

60. The orbitals present in the anti-penultimate shell can't be involved in hybridization.

61. Ionic conductance increases with increasing the number of ions, produced after decomposition.

62. M.P order is $C > Si > Ge > Pb > Sn$

Covalent radius order is $Pb > Sn > Ge > Si > C$

63. Aniline is Steam volatile

Camphor shows sublimation

H_2O_2 decomposes below its boiling point

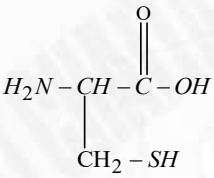
Crude oil purified by fractional distillation

64. Ketones react with Grignard reagent to give tertiary alcohols.

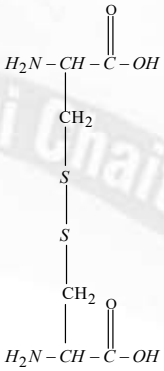
65. Alcyl group is meta directing group

66. Aldehyde and ketones react with 2,4-DNP, alcohols reacts with Na

67.

Cysteine 

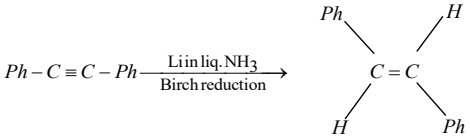
Cystine



68. A, C, D Molecules are symmetric

69. Refer N C E R T 12 pg 224

70. With Lithium in liquid ammonia, trans-alkene is almost an exclusive product



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71. B.O of O_2, O_2^-, O_2^+ and O_2^{-2} are $2, 1.5, 2.5 \& 1$

72. $\text{Charge passed} = \frac{8\times20\times0.5\times60}{96{,}500} = 0.05\text{F}$ $\quad\therefore$ net metal $= 0.05 = \frac{5.2}{\text{GEW}}$

$\therefore \text{GEW} = \frac{5.2}{0.05} = 104$

73. Here, $\lambda = \frac{h}{\sqrt{2\times m\times eV}} = \frac{6.63\times10^{-34}}{\sqrt{2\times1.6\times10^{-19}\times40000\times9.1\times10^{-31}}}m$

$\therefore \lambda = 0.614\times10^{-11}m$

$\lambda = 6.14\times10^{12}m$ [Nearest integer = 6]

74. $Na_2[Fe(CN)_5NO] + Na_2S \rightarrow Na_4[Fe(CN)_5NOS]$ (no change in oxidation state of Fe)

75. Glycogen is also called as animal starch.

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